Fanteng Meng

Ph.D. Student in Safety Science & Engineering

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1 Education

2023 – Present	**Ph.D., Safety Science & Engineering** Beijing Jiaotong University (BJTU), State Key Lab of Rail Traffic Control & Safety & Transportation College (Intelligent Systems Dept.) Advisor: Prof. Yong Qin Enrolled via successive master-doctor program since 2023.
2022 – 2023	**M.Eng., Transportation Planning & Management** Beijing Jiaotong University (BJTU)
2018 – 2022	**B.Eng., Transportation** Shijiazhuang Tiedao University

2 Research Interests

UAV-based intelligent inspection; Machine vision & deep learning for rail transit; Edge computing; 3D reconstruction & point cloud analysis; Full-stack low-altitude intelligent perception systems.

3 Research Projects

P.I. for the item marked (P.I.); the rest are core participants (Core).

1. **Development and Demonstration Verification of Safety-Guarantee Technologies for Autonomous Rail Transit Systems.** (2022–2026) – National Key R&D Program (Core)
2. **Research on Unknown-Risk Identification of High-Speed Railway Surroundings from UAV Remote Sensing Imagery.** (2024–2026) – Fundamental Research Funds for the Central Universities (P.I.)
3. **In-depth Research and Application of UAV Inspection Technology for High-Speed Railway Infrastructure.** (2022–2024) – China Railway "Major Project" (Core)
4. **Construction of an Automatic UAV Intelligent Inspection System for Beijing–Shanghai HSR Infrastructure.** (2023–2024) – Beijing–Shanghai HSR "Major Project" (Core)
5. **Research on Automated Overhaul Equipment and Key Technologies for Railway Steel-Truss-Bridge Coating.** (2024–2027) – Beijing–Shanghai HSR "Major Project" (Core)
6. **Application of Autonomous-UAV Inspection to Defects and Rockfall Debris in Key Zones of Qinghai–Tibet Railway Bridges.** (2026–2027) – "Smart Sky Road" Major Project of the Qinghai–Tibet Railway (Core)

4 Selected Publications

* *Corresponding author* $^\#$ *Co-first author*

Journal Articles

1. **Meng, F.**, Qin, Y.*, Wu, Y.*, et al. "SRLF: Sparse Representation Learning Framework for Railroad Surrounding Potential Risk Perception Using UAV Imagery." *IEEE Trans. Intell. Transp. Syst.*, 2025.
2. **Meng, F.**, Qin, Y.*, Wu, Y.*, Shao, C., Yang, H., Jia, L. "Automatic Risk Level Evaluation System for Potential Environmental Hazards along High-speed Railroad Using UAV Aerial Photograph." *Expert Syst. Appl.*, 2025.
3. **Meng, F.**, Qin, Y.*, Wu, Y.*, Shao, C., Jia, L. "A Subtle Defect Recognition Method for Catenary Fastener in High-speed Railroad Using Destruction and Reconstruction Learning." *Adv. Eng. Inform.*, 2024.
4. Wu, Y.#, **Meng, F.#**, Qin, Y.*, Qian, Y.*, Xu, F., Jia, L. "UAV Imagery Based Potential Safety Hazard Evaluation for High-speed Railroad Using Real-time Instance Segmentation." *Adv. Eng. Inform.*, 2023.
5. Wu, Y.#, **Meng, F.#**, Qin, Y.*, Qian, Y.*, Liu, Z., Zhao, W. "Automated Anomaly Detection of Catenary Split Pins Using Unsupervised Learning." *Autom. Constr.*, 2024.
6. **Meng, F.**, Qin, Y.*, Cui, J., Wu, Y., Zhang, Z., Wei, S. "Unknown Risk Detection in External Environment of Railroad Using UAV Images." *Acta Aeronaut. Astronaut. Sin.*, 2025.
7. Qin, Y.*, **Meng, F.#**, Zhang, Z., et al. "Autonomous UAV-based Intelligent Inspection Framework for Rail Transit Infrastructure: A Review." *J. Beijing Jiaotong Univ.*, 2025.

5 Selected Engineering Projects

1. **Fully-automatic UAV Intelligent Inspection System for High-speed Railway**
   - Core researcher in national key R&D program; designed 3 vision algorithms (>96% recall).
   - Deployed 6 UAVs + 4 auto drone nests; achieved 10× efficiency gain over manual inspection.
2. **UAV Inspection for Bridges & Rockfall Hazards, Qinghai-Tibet Railway**
   - Full-stack R&D: autonomous flight control, photogrammetry 3D reconstruction, point cloud segmentation.
   - Multi-temporal registration & rockfall deformation detection for plateau railway safety.
3. **Full-coverage UAV Inspection & Coating Defect Evaluation for Railway Steel Truss Bridge**
   - Designed obstacle-avoidance flight modes covering top, side, & under-bridge zones.
   - End-to-end workflow: coating defect recognition → quantitative grading → spatial traceability.

6 Patents

Granted Patents

1. Autonomous UAV Based Intelligent Inspection System for Equipment, Facilities, and the Environment along Railway Lines and Method Thereof. (US Patent)
2. Unknown Risk Identification Method for Railway Surrounding Environment Using UAV Remote Sensing Images.
3. UAV Route Generation Method for Transportation Facility Inspection under GNSS-denied Environments.
4. Cross-modal Adaptive Matching Based Online Extrinsic Calibration Method and System for LiDAR and Camera.

Published Patent Applications

1. Refined Catenary Split Pin Defect Recognition Method for Railway Catenary.
2. Coating Corrosion Detection, Assessment Method and System for Railway Steel Truss Bridges.

3. Railway Environmental Hazard Identification and Risk Level Assessment Method Based on UAV Images.

7 Honors & Awards

2024 Leading Technology Award (Outstanding Achievement), World Internet Conference — "Fully-autonomous UAV Intelligent Inspection Technology and Application for Rail Transit"

2024 Silver Award, 14th Challenge Cup National College Students' Entrepreneurship Competition (Belt & Road Intl. Invitational)

2024 Silver Award, "Qingchuang Beijing" Capital College Students Challenge Cup Entrepreneurship Competition

2022 Second Place, Beijing Jiaotong University Summer Deep Learning Championship

2021 First Prize, China Transportation Outstanding Student Award

2021 Provincial First Prize, 7th National College Students Engineering Training Integration Ability Competition

2021 Provincial Merit Student

2020 Provincial First Prize, 12th National College Students Mathematics Competition

2020 Provincial First Prize, 7th College Students Physics Competition

8 Professional Service

Journal Reviewer *IEEE Trans. Intell. Transp. Syst.* (T-ITS), *Adv. Eng. Inform.* (AEI), *Expert Syst. Appl.* (ESWA)

9 Technical Skills

Languages: Python, C++, Matlab **Deep Learning:** PyTorch **Systems:** Linux, ROS **Vision:** OpenCV **Hardware:** DJI-PSDK